

# Main Controller Guide (v2.0.0 – 2025-10-16)

## Overview

The Main Controller coordinates up to six Station Controllers (IDs 0–5) over Ethernet using the W5500 module. It manages relay states, monitors inputs, provides LED feedback, and performs station overrides.

## Pin Assignments

| Function           | Pins        | Notes                                   |
|--------------------|-------------|-----------------------------------------|
| Switch Inputs (21) | D2–D21 + D1 | D1 used only when SERIAL_DEBUG = false  |
| LED Outputs (42)   | D24–D69     | Even = LED_A (Red), Odd = LED_B (Green) |
| Ethernet CS        | D22         | W5500 Chip Select                       |
| Ethernet RESET     | D23         | Reset pin                               |
| SPI (Ethernet)     | D50–D53     | MOSI=51, MISO=50, SCK=52, SS=53 (HIGH)  |
| MCP23017           | Removed     | Not used                                |

## Override Logic

Station 0 input A3 forces ON relays tied to inputs 4,5,11,16. Station 4 input A4 forces ON relay tied to input 19. Overrides take precedence until released.